



Simultaneous realization of antifouling, self-healing, and strong substrate adhesion via a bioinspired self-stratification strategy

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ABSTRACT

Antifouling ability and substrate adhesion are typical trade-offs seen for a silicone coating. Furthermore, the self-healable inability limits their duration. This study reports a bioinspired self-stratifying coating composed of a polyurethane with dopamine and silane pendant groups and a bi-silanol-terminated poly(dimethylsiloxane) with fluorocarbon and poly(ethylene glycol) segments as a reactive amphiphilic polymer (RAP). During coating formation, the RAP with low surface energy can be enriched on the surface due to the self-stratifying effects, whereas the bioinspired dopamine and silane groups provide the coating with strong substrate adhesion. Besides, the multiple hydrogen bonds formed between the urethane linkages and the dopamine groups allow the coating to heal. Our study indicates such a polyurethane-based coating has excellent antifouling ability against proteins, bacteria, and diatoms, good self-healing performance at room temperature in air and water, as well as high adhesion strength on various substrates such as ceramic, steel, glass, and wood. This study provides a promising and universal strategy for the simultaneous realization of antifouling, self-healing, and strong substrate adhesion. The coatings could meet the needs of applications in marine antifouling, biomedical instruments, and other fields.

1. Introduction

Deposition of unwanted microorganisms onto surfaces generally results in biofouling that can cause serious impacts on marine facilities, biomedical instruments, and other applications [1–3]. Given the severity of the consequences, there is a growing interest understanding and reducing fouling. Coating the surfaces with antifouling materials may be the most direct and effective way to inhibit biofouling [4–6].

Generally, traditional antifouling coatings contain toxic biocides that may hurt the ecology [7]. Ecofriendly antifouling coatings, such as silicone-based fouling release coatings, have been explored to inhibit biofouling [8]. Poly(dimethylsiloxane) (PDMS) elastomers are widely examined, which can weaken the adhesion of fouling organisms to surfaces due to their low surface energy and low elastic modulus [9,10]. However, they generally exhibit poor adhesion strength to substrates [11]. Actually, the antifouling ability and substrate adhesion are mutually exclusive for a silicone coating. Urethane and epoxy groups are frequently introduced into PDMS elastomers to improve adhesion strength at the expense of antifouling properties due to surface changes caused by those polar groups [12,13]. Modifying the PDMS elastomer

with fouling resistant groups, such as poly(ethylene glycol) (PEG) or zwitterionic polymers, may be a feasible approach to reduce biofouling [14]. PEG-based or zwitterionic polymers can bind water strongly to form a physical energy barrier that keeps biomolecules and microorganisms away from the surfaces [2,15]. However, the high hydrophilicity of these coatings may sacrifice the bulk mechanical properties and adhesion strength, since they have high surface free energy and often require the incorporation of a large amount of PEG or zwitterionic moieties [16]. Several studies have been shown that having more than 30% of zwitterionic or PEG groups in a coating has adverse impacts on the overall mechanical properties [17–21]. It is vital to develop a method to achieve effective partitioning that results in an outer surface displaying purely antifouling moieties, while the body guaranteeing mechanical properties and adhesion strength. Our group have reported the PDMS based coating with surface-enriched PEG or zwitterions by the self-stratifying effects [22,23]. However, because they are fixed in the cross-linking sites, high PEG or zwitterions contents are required to achieve the desired antifouling performance. Besides, the coating with low modulus easily suffered from mechanical damage and cannot heal due to its crosslinked structure [24]. It is a significant challenge to

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achieve the antifouling, self-healing, and strong substrate adhesion at the same time.

Marine mussels exhibit a unique ability to fix themselves to various surfaces even in a wet environment, which is caused by the adhesive proteins with high interfacial binding durability [25–27]. Through hydrogen bonding, π - π aromatic interactions, and metal-ligand complexation, 3,4-dihydroxyphenylalanine plays an important role in curing and interfacial interactions [25]. Dopamine moieties have been incorporated into numerous coating systems to improve the adhesion strength [28–30]. Moreover, it can also enable the material with self-healing properties [31–33].

In this study, we prepared a bioinspired self-stratifying coating consisting of polyurethane with dopamine and silane pendant groups and a bi-silanol-terminated PDMS with fluorocarbon and poly(ethylene glycol) segments as a reactive amphiphilic polymer (RAP). First, the polyurethane was prepared using polytetramethylene glycol and isophorone diisocyanate as the soft and hard segments, respectively, and a dopamine-derived diol, a silane-derived diol, and 1,4-butanediol as the chain extender. The RAP was prepared by radical telomerization reaction followed by a condensation reaction, which was crosslinked into the polyurethane by the hydrolysis and condensation reactions between the silanol groups of RAP and the pendant silane groups of PU. Because of its low surface energy, RAP segregates to the surface of the coating, while polyurethane forms the bulk of the coating system during film formation. This self-stratifying effect gives rise to an amphiphilic surface with good antifouling ability toward proteins, bacteria, and diatoms, while the dopamine and silane groups provide the coating with strong substrate adhesion toward diverse surfaces, even in wet and underwater environments. Furthermore, the multiple hydrogen bonds formed by the urethane linkages and dopamine groups allow the coating to heal at room temperature. This method achieves antifouling moieties on the outer surface, high adhesion strength on the bottom, and self-healing properties on the main body. We aim to develop a bioinspired self-stratification strategy to simultaneously achieve the antifouling, self-healing, and strong substrate adhesion for the potential applications in marine antifouling, biomedical instruments, and other fields.

2. Experimental section

2.1. Materials

3-(Dimethoxy(methyl)silyl)-propane-1-thiol, poly(ethylene glycol) methyl ether methacrylate ($M_n = 475$ g/mol), dodecafluoroheptyl methacrylate, dichlorodimethylsilane, 3-mercaptop-1,2-propanediol (TPG), 3-(trimethoxysilyl)propylmethacrylate (MPS), p-toluenesulfonic acid, isophorone diisocyanate (IPDI), 1,4-butanediol, dibutyltin dilaurate, dopamine hydrochloride (DA), trifluoroacetic acid, 2,2-Bis(hydroxymethyl)propionic acid (DMPA), benzil dimethylketal, 2,2-dimethoxypropane (DMP), N-Hydroxysuccinimide (NHS), N-(3-dimethylaminopropyl)-N'-ethylcarbodiimide hydrochloride (EDC-HCl), triethylamine were purchased from Aladdin and used as received. Azobis(isobutyronitrile) (AIBN, Aladdin) was purified by recrystallization from methanol. Polytetramethylene glycol (PTMG, $M_n = 1000$ g/mol) from Aladdin was dried under reduced pressure at 100 °C for 2 h. Fibrinogen (fraction I from human plasma, $M_w = 340$ kDa, pI 5.5) was from Merck Chemicals. Dichloromethane, methanol, acetone, and hexane were purchased from Sinopharm and used as received. Tetrahydrofuran (THF) and N,N-Dimethylformamide (DMF) from Sinopharm were distilled from CaH₂ prior to use. Artificial seawater (ASW) was prepared following ASTM D1141-98 (2003).

2.2. Synthesis of amphiphilic telomer

3-(Dimethoxy(methyl)silyl)-propane-1-thiol (10 mmol, 1.803 g), poly(ethylene glycol) methyl ether methacrylate (34.2 mmol, 16.245 g), dodecafluoroheptyl methacrylate (17.1 mmol, 6.84 g), and AIBN (0.059

g) were dissolved in THF (40 mL) and introduced into a flask with a magnetic stirring bar. The solution was bubbling by argon for 1 h and then stirred at 70 °C for 24 h. The resulting solution was precipitated in hexane and a colorless transparent viscous liquid was obtained. ¹H NMR (600 MHz, CDCl₃, Fig. 1a, ppm): 5.44 (CF₂CHF₂), 4.50 (–COOCH₂CF₂), 4.05 (–COOCH₂CH₂O–), 3.70 (COOCH₂CH₂O), 3.42 (SiOCH₃), 3.32 (CH₂CH₂OCH₃), 2.44 (–SiCH₂CH₂CH₂S), 1.53 (SiCH₂CH₂CH₂), 0.58 (SiCH₂CH₂CH₂), 0.10 (SiCH₃). The molar mass was calculated to be ~3000 g/mol according to the ¹H NMR.

2.3. Synthesis of bis-silanol terminated siloxane with telomer side chains (RAP)

RAP was synthesized by the hydrolysis of the amphiphilic telomer (Scheme 1a). Telomer (4.0 g) in 10 mL THF and dichlorodimethylsilane (17.336 g) were added dropwise into 50 mL DI water in three-neck flask. After stirring at 80 °C for 3 h, saturated NaHCO₃ solution was added to neutralize the HCl generated during the reaction. The resulting liquid was extracted by ethyl acetate and the ethyl acetate layer was dried using Na₂SO₄, filtered and concentrated to give opaque oily product. The product was dialysed and finally the mixture was dried to give final product. ¹H NMR (600 MHz, CDCl₃, Fig. S1, ppm): 5.46 (CF₂CHF₂), 4.57 (COOCH₂CF₂), 4.07 (COOCH₂CH₂O), 3.70 (COOCH₂CH₂O), 3.37 (CH₂CH₂OCH₃), 2.44 (SiCH₂CH₂CH₂S), 1.53 (SiCH₂CH₂CH₂), 0.53 (SiCH₂CH₂CH₂), 0.06 (SiCH₃). The number average molecular weight (M_n) of RAP is 6200 and polydispersity (PDI) is 1.59.

2.4. Synthesis of silane-derived diol chain extender (TPG-MPS)

TPG-MPS was synthesized by a thiol-ene click reaction (Scheme S1). Typically, TPG (21 mmol) and MPS (20 mmol), and benzil dimethylketal (0.02 g) were dissolved in 25 mL THF. The solution was degassed with nitrogen for 30 min. The sample was incubated by irradiation with a UV-lamp (emitting nominally at 350 nm, light intensity 100%, 100 mW/cm²) for 40 min. The product was precipitated into water, filtered, and dried under vacuum at room temperature for 24 h. ¹H NMR (600 MHz, CDCl₃, Fig. S2, ppm): 3.47 (–SiOCH₃), 2.68 (CH₂SCH₂CH(CH₃)), 2.47 (CH₂CH(CH₃)CO) 1.73 (OCH₂CH₂CH₂Si), 0.58–0.72 (OCH₂CH₂CH₂Si).

2.5. Synthesis of DMP-DMPA (Scheme S2)

DMPA (26.8 g, 200 mmol) and DMP (27.0 g, 260 mmol) were dissolved in 200 mL acetone. Then p-toluenesulfonic acid (0.2 g, 1.06 mmol) was added into the solution at room temperature. The solution was stirred for 5 h, and then NaHCO₃ (10.5 g, 100 mmol) was added to neutralize the solution. The solution was stirred for another 20 min, filtered, and concentrated to obtain the crude product. The crude product was dissolved in 150 mL dichloromethane, followed by washing with deionized water (30 mL) 3 times. After drying over anhydrous Na₂SO₄, the organic solution was concentrated by rotary evaporator. Finally, the product was dried under 60 °C for 12 h. The obtained DMP-DMPA was a white solid with a yield of 92%. ¹H NMR (600 MHz, CDCl₃, ppm, Fig. S3) δ 12.50 (s, 1H), 4.00 (d, 2H), 3.56 (d, 2H), 1.34–1.27 (d, 6H), 1.07 (s, 3H).

2.6. Synthesis of DMP-DMPA-NHS

DMP-DMPA (8.7 g, 50 mmol) and NHS (6.85 g, 60 mmol) were added into a 250 mL single-necked flask, and then 200 mL dichloromethane was dissolved the mixture. EDC-HCl (23.0 g, 120 mmol) was added into the solution and stirred at 0 °C for 2 h and 24 h at room temperature. The crude product was washed with saturated NaHCO₃ solution, dilute hydrochloric acid solution and water for 3 times, respectively. The organic phase was dried by anhydrous Na₂SO₄, filtered, and concentrated to obtain white solid. Finally, the product was dried under 60 °C for 12 h in vacuum to obtain DMP-DMPA-NHS with a yield of 91%. ¹H NMR (600

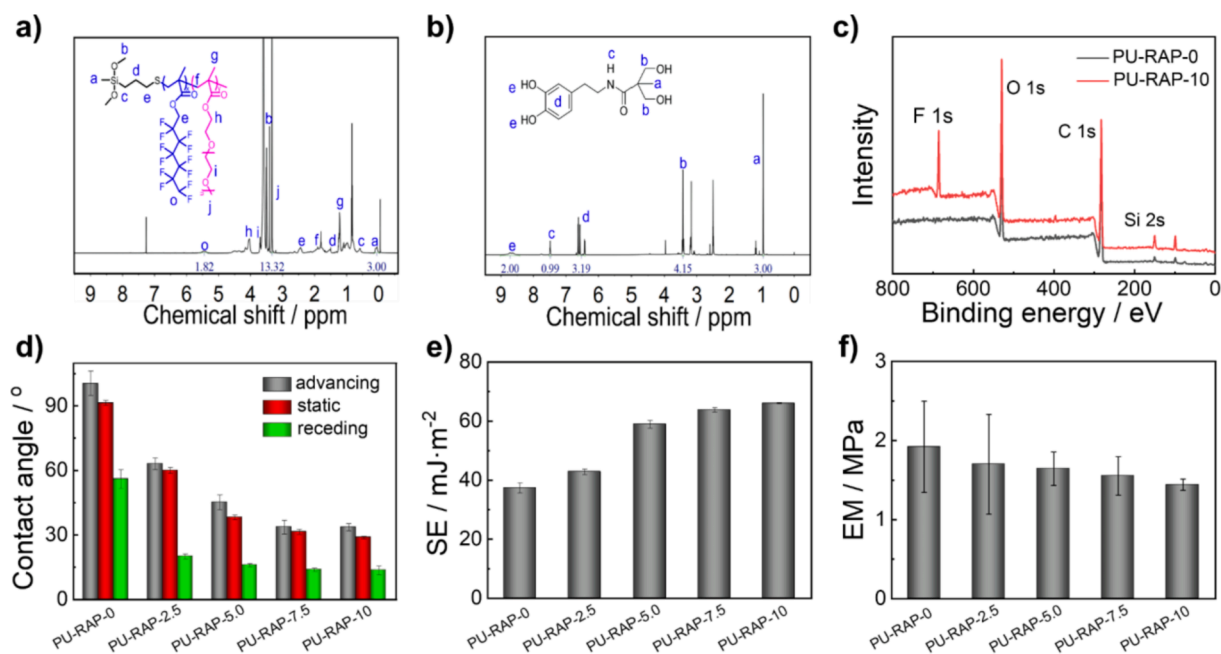
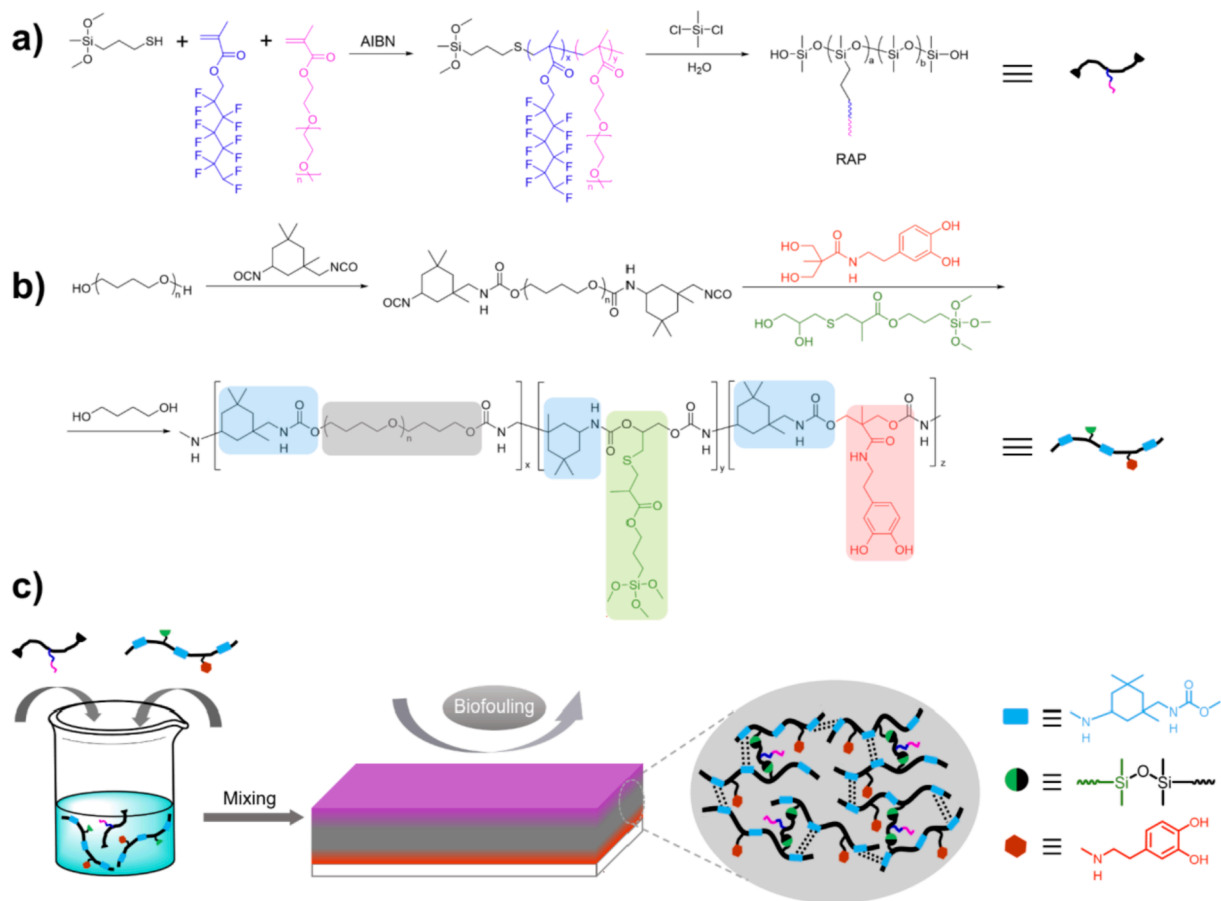


Fig. 1. ¹H NMR spectra of the telomer (a) and DMPA-DA (b). (c) XPS spectra of the PU-RAP-0 and PU-RAP-10. (d) Advancing, static and receding water contact angle of PU-RAP-x coatings. (e) Surface energy (SE) and (f) elastic modulus (EM) of PU-RAP-x coatings.



Scheme 1. (a) Preparation of the reactive amphiphilic polymer (RAP). (b) Synthesis of polyurethane with dopamine and silane pendant groups. (c) Schematic diagram of the preparation of PU-RAP-x coating.

MHz, CDCl_3 , ppm, Fig. S4) δ 4.08 (d, 2H), 3.81 (d, 2H), 2.81 (s, 4H), 1.40–1.29 (d, 6H), 1.21 (s, 3H).

2.7. Synthesis of DMP-DMPA-DA

DMP-DMPA-NHS (5.43 g, 20 mmol) was added to a 250 mL three-necked flask under nitrogen atmosphere. Then 70 mL dichloromethane and 70 mL methanol were added into the three-necked flask. After the DMP-DMPA-NHS was totally dissolved, DA-HCl (5.69 g, 30 mmol) and triethylamine (3.03 g, 30 mmol) were then added and stirred at 0 °C for 24 h under the nitrogen atmosphere. The solution was concentrated and precipitated in 1 M HCl aqueous solution. The product was filtered and dried under 60 °C for 24 h. The yield was 87%. ^1H NMR (600 MHz, CDCl_3 , ppm, Fig. S5) δ 8.73–8.61 (d, 2H), 7.51 (s, 1H), 6.64–6.41 (m, 3H), 3.92 (d, 2H), 3.56 (d, 2H), 1.32–1.24 (d, 6H), 1.03 (s, 3H).

2.8. Synthesis of DMPA-DA

DMP-DMPA-DA (6.18 g, 20 mmol) was dissolved in 130 mL methanol. Trifluoroacetic acid (2.3 g) was added under nitrogen atmosphere and the reaction was stirred for 12 h at room temperature. Then, the solvent was removed under reduced pressure and the product was dried under vacuum oven for 24 h. The yield was about 90%. ^1H NMR (600 MHz, CDCl_3 , ppm, Fig. 1b) δ 8.83–8.56 (d, 2H), 7.48 (s, 1H), 6.61–6.42 (m, 3H), 3.43 (s, 4H), 0.94 (s, 3H).

2.9. Synthesis of polyurethane with pendent catechol and silane groups

IPDI (14 mmol), PTMG (4 mmol), DMF (60 mL), and dibutyltin dilaurate (0.2 wt%) were added into three-necked flask equipped with condenser, nitrogen gas inlet, and magnetic stirrer bar. The mixture was stirred under 75 °C for 4 h. Then, DMPA-DA (3 mmol) was added into the mixture and stirred at 85 °C for 2 h. The TPG-MPS (3 mmol) and 1,4-butanediol (4 mmol) were added and stirred at 85 °C for another 3 h. The product was precipitated into water, filtered, and dried under vacuum at room temperature for 24 h after the DMF was removed by rotary evaporator. The M_n of PU is 9200 and PDI is 1.52.

2.10. Coating preparation

The coating was prepared by condensation reaction between the RAP and the PU. Different contents of RAP were mixed with PU. Typically, PU (2.5 g), RAP (0.132 g) and catalyst dibutyltin dilaurate (13 mg) were dissolved THF or chloroform and stirred with a magnetic bar for 1 h. The solution was cast onto glass slides (for surface wettability, antibacterial assays, and anti-diatom test) and spin-cast on AT-cut quartz crystal surface (for protein adsorption test). The coating was placed into an enclosed drying cabinet with ventilation for 12 h at room temperature followed by curing at 60 °C for another 12 h. The coating is designated as PU-RAP-x, where x refers to the mass percentage of RAP in the coating.

2.11. Characterization

Nuclear Magnetic Resonance Spectroscopy (^1H NMR) analysis was recorded on a Bruker AV600 NMR spectrometer using CDCl_3 or $\text{DMSO}-d_6$ as the solvent and tetramethylsilane (TMS) as the interior label. Gel permeation chromatography (GPC) was carried out on Agilent 1260 Infinity with refractive index and UV detectors. The test was performed in THF (flow rate: 1.0 mL/min) at 35 °C with two identical PLgel columns (5 μm , MIXED-C). Calibration curve was determined by pretests on polystyrene standards (1.30×10^3 to 2.21×10^6 g/mol) with narrow polydispersity. X-ray Photoelectron Spectroscopy (XPS) analysis was performed using an Axis Ultra DLD (Kratos Analytical) with a monochromate Al K α ($h\nu = 1486.6$ eV) source at takeoff angles of 90° to investigate the surface compositions. Water Contact Angle (WCA) and

Surface Energy (SE) measurements were conducted on a theta optical tensiometer (Attension, Biolin Scientific) by depositing 3 μL water or diiodomethane droplets on the coating surface using the sessile method at 25 °C. The average of the results from five measurements was used as the final CA value. The SE was calculated from the WCA of five measurements and the diiodomethane contact angle of five measurements using the Owens-Wendt-Rabel-Kaelble method. The surface elastic modulus measurements were conducted with an XE-100 atomic force microscopy (AFM) measurements (Park Systems) by applying Hertzian contact theory, which correlates the AFM force signal (F) to the indentation depth of the cantilever tip (δ) and the elastic modulus (E) using the following equation:

$$F = \frac{E}{1 - \nu^2} \frac{\tan \alpha}{\sqrt{2}} \delta^2$$

where ν is Poisson's ratio (0.5 for all coatings), α is the face angle of the four-sided pyramidal tip of the cantilever (22°), and δ is the indentation depth.

2.12. Adhesion test

The adhesion strength of PU-RAP-x coatings to different substrates (glass fiber reinforced epoxy panel (EP), ceramic, steel, glass, wood, and plastic) was measured by using an adhesion tester (PosiTest AT-A Automatic) according to the ASTM D4541. An aluminum dolly was adhered to the surface of the coatings using epoxy adhesive (Araldite Rapid Epoxy Adhesive, Huntsman). The data were gathered by detaching the dolly at a speed of 0.2 MPa s^{-1} . An average value was calculated from five different test points on each coating. The wet surface was prepared by covering 100 μL water onto EP (7.5 cm $\times$ 2.5 cm). Underwater adhesion test: the coating on aluminum dolly was adhered on different underwater surfaces and pressed with a weight of 2 kg for 24 h.

2.13. Self-healing test

The macroscopic self-healing test of the polymer film in air or water was recorded using an optical microscope at room temperature.

2.14. Tensile test

The mechanical properties and self-healing efficiency of PU-RAP-x coatings were evaluated by a tensile test using an Instron 5966 instrument with a 100 N load cell at 25 °C. The sample was rectangular shaped with 10 mm width, 0.5 mm thickness and 50 mm gage length. The rate of elongation was 25 mm min^{-1} and all the tests were repeated at least three times. For evaluating self-healing efficiency, elastomer films were cut using a razor blade, and then reconnected quickly and allowed to heal at room temperature.

2.15. Quartz crystal microbalance with dissipation (QCM-D)

An El system (Q-sense AB, Sweden) was used to conduct the QCM-D measurements. The polymer solution in THF (5 mg/mL) was spin cast on AT-cut quartz crystal surface and cured at 60 °C for 12 h. ASW was used as reference, and 1 mg/mL protein solutions was delivered to the surface at a flow rate of 150 $\mu\text{L/min}$. Details of the QCM-D measurements can be found elsewhere. Briefly, the change of frequency (Δf) is related to the protein adsorption, whereas the change of dissipation (ΔD) is correlated with the viscoelastic properties of the adsorbed protein layer.

2.16. Antibacterial assays

The antibacterial performance was evaluated using the Gram-positive *S. aureus*, Gram-negative *E. coli*. The sterilized PU-RAP-x coatings (8 mm diameter) were put in a 48-well cell-culture plate, and they

were added by 100 μL bacterial suspension (10^6 cells/mL). After incubation at 37°C for 4 h, their surfaces were gently rinsed three times with 1 mL of PBS. 100 μL of the last washing fluid using an ultrasonic cleaner was dispersed in the broth and cultured for at 37°C for 24 h. We also evaluated the antibacterial performance using fluorescent method and use the marine bacteria *Pseudomonas. sp nov 776* as an example. PU-RAP-x coatings were immersed in 1 mL bacterial suspension of ASW (10^7 cells/mL) at 30°C for 5 h. Subsequently, the coating was gently washed with ASW to remove the unattached bacteria and then stained with the LIVE/DEAD BacLight Bacterial Viability Kit for 0.5 h. The adhered cells were examined under a fluorescence microscope (Scope A1, Zeiss). The number of bacteria was determined from ten random fields of view on triplicate coatings.

2.17. Anti-diatom test

The diatom *Navicula incerta* (*N. incerta*) was obtained from Xiamen University. It was cultivated in F/2 culture medium under a 12:12 h light–dark cycle at 23°C in a light incubator. A 1.0 mL sample of diatom suspension (a chlorophyll value of 0.25 $\mu\text{g/L}$) was added to a 24-well microplate and the film was immersed in the diatom suspension. After culturing for 24 h, the film was gently washed with ASW to remove the unattached diatoms. Adhered diatoms were observed under a microscope, and counts were made for ten fields of view on each of three replicate slides, and the average value of the density of the attached diatoms was recorded.

3. Results and discussion

Scheme 1 illustrates the preparation of the RAP and the polyurethane (PU). First, the telomer was prepared through free radical polymerization with 3-(dimethoxy(methyl)silyl)-propane-1-thiol as a chain transfer agent and the RAP was synthesized by hydrolysis and condensation reactions of the telomer and dichlorodimethylsilane. Fig. 1a shows the ^1H NMR of telomer. Peaks at 0.20 ppm (peak a) and 3.69 ppm (peak b) are attributed to the protons of SiCH_3 and SiOCH_3 in the silane. The peak at 5.47 ppm (peak o) can be assigned to the protons of fluorocarbon groups. Peaks at 3.39 (peak j), 3.61 (peak i), and 4.07 ppm (peak h) can be assigned to the protons of CH_2 and CH_3O in the PEG moiety and CH_2 linked to the ester groups, respectively. These facts clearly indicate the successful synthesis of the telomer. According to the integral result, the molar mass of the telomer was calculated to be ~ 3000 g/mol. By the co-hydrolysis of the telomer and dichlorodimethylsilane, it can form RAP which is supported by the fact that peaks of fluorocarbon and PEG groups appear and the peak of SiOCH_3 disappear in the ^1H NMR of RAP (Fig. S1). Next, the PU was synthesized following a one-pot two-step protocol: formation of the pre-polymer and its chain extension. To obtain PUs with silane and pendent catechol groups, three chain extenders were used: dopamine-derived diol (DMPA-DA), silane-derived diol (TPG-MPS), and 1,4-butanediol. The synthetic routes and ^1H NMR of DMPA-DA and TPG-MPS were shown in Scheme S1–S2, Fig. 1b and Fig. S2–S5. As shown in Fig. S6, peaks at 8.61 and 8.67 ppm can be assigned to the protons of phenolic groups in the dopamine. The peak at 3.3 ppm is assigned to silane from the TPG-MPS. These facts indicate that TPG-MPS and DMPA-DA are successfully participated in polymerization. Finally, the RAP can crosslink directly with the PU by the hydrolysis and condensation reactions between the silanol groups of RAP and the pendant silane groups of PU. Four different RAP contents (2.5%, 5.0%, 7.5%, and 10%) were prepared, and the coating formulations are shown in Table S1. For convenience, the hybrid coatings were termed PU-RAP-x, where x refers to the content of RAP.

During film formation, the RAP can self-stratify to the surface of the coating due to the low surface energy of silicone and fluorocarbon segment of RAP and its incompatibility with PU [22,34]. To verify the self-stratifying properties of the RAP, the XPS was used to study the surface composition of the coatings. As exhibited in Fig. 1c, the peak at

686 eV belongs to F (1s) in the fluorocarbon groups of PU-RAP-10. For PU-RAP-0, no F (1s) can be observed. More importantly, the F content of the PU-RAP-10 coating is lower in the bulk than on the surface (Table S2), indicating that the low surface energy of fluorocarbon groups is the driving force for the migration of RAP to the surface. In this case, PEG moiety is driven upward to the surface because PEG and fluorocarbon segments are randomly distributed in the RAP side chain [22].

The self-stratifying properties of the RAP were also examined by measuring the water contact angle (WCA) of PU-RAP-x coatings. Fig. 1d shows that the static WCA of PU-RAP-x coatings decreased from $\sim 90^\circ$ to $\sim 30^\circ$ with the content of RAP. This is due to an increase in the amount of hydrophilic PEG on the surface. The dynamic WCA follows a similar trend to the static WCA. In general, the advancing WCA is related to the hydrophobic character of the surface, whereas the receding WCA is an estimate of the hydrophilic character of the surface [35]. The values of advancing WCA are slightly higher than the static WCA. In contrast, the receding WCA is lower than the static WCA. The receding WCA of a coating containing RAP with silicone, fluorocarbon, and PEG moieties, decreases as the PEG content increases, clearly indicating that the PEG is driven upward to the surface of the coating by the linked fluorocarbon groups. The surface energy (SE) of the PU-RAP-x coatings was also examined (Fig. 1e). SE for the coatings was calculated using WCA and diiodomethane contact angles (Table S3). The SE value for the PU-RAP-0 coating is ~ 38 mJ m^{-2} and increases with the RAP content. This increase correlates with the WCA. The greater the change in WCA values, the greater the change in the SE of a coating [36]. The surface modulus was also examined because a lower modulus is favorable for fouling release. The PU has a modulus of ~ 2.0 MPa, and the surface modulus of PU-RAP-x coatings decreases as RAP content increases (Fig. 1f).

The mechanical properties of PU-RAP-x coatings by a tensile test were shown in Fig. 2a. We found that the mechanical properties of different PU-RAP-x coatings show a similar tensile strength when varying the RAP content, which can be stretched up to a high strain (Fig. 2b). When polyurethane is used as a coating, it is unavoidable that damage will occur during practical use. Thus, a good self-healing property can extend the lifetime of the coating. The self-healing property of PU-RAP-x coatings was examined by using optical microscopy, as observed in Fig. 2c. The sample was cut with a razor blade and placed at room temperature (25°C) without any further treatment. After 24 h, the crack in the PU-RAP-x coatings gradually healed and vanished. To show it more intuitively, we cut the PU-RAP-5.0 sample into two pieces and joined them together for self-healing at 25°C for 24 h, which could remain high stretchability (Fig. 2d). Moreover, it can also self-heal underwater. The crack almost disappeared after healing underwater at 25°C for 24 h (Fig. S7). The tensile tests of PU-RAP-5.0 sample before and after healed at 25°C for 24 h was conducted to calculate the self-healing efficiency. As shown in Fig. S8, the self-healing efficiency of PU-RAP-5.0 reached $\sim 98\%$, when the strain was taken into account, which was better than our previous reported amphiphilic polyurethane coating [37]. Two aspects can be attributed to self-healing ability [26,31,38]. First, polymer chain mobility and diffusion are critical in the crack closure [38]. Second, dynamic hydrogen bonds can also help the system heal itself. In such system, higher content of hydrogen bonds between the N–H and C=O groups from the urethane linkages and phenol groups of dopamine makes a great contribution to self-healing [26].

Dopamine-functionalized polymers were reported to have excellent adhesion performance due to the formation of covalent bonds and physical interactions with the substrates [28,39,40]. We characterized the AS of PU-RAP-x coatings on a dry surface, wet surface, and underwater surface using a glass-fiber reinforced epoxy panel (EP) as a substrate. As shown in Fig. 3a, the AS of PU-RAP-x coatings slightly decrease with the addition of the RAP content, and PU-RAP-0 exhibits the highest AS of ~ 1.5 MPa.

The hydrogen bonds from the urethane groups and polar bonds from

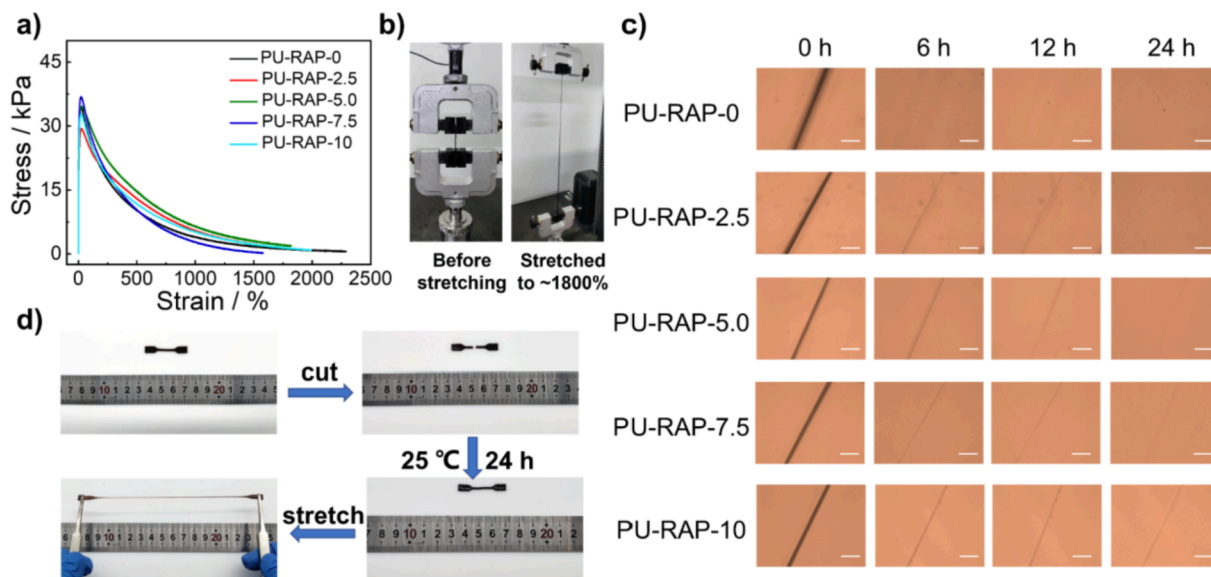


Fig. 2. (a) Stress-strain curves of the PU-RAP-x coating. (b) Photos of the high stretchability of the PU-RAP-5.0 sample before and after stretched. (c) Self-healing process of PU-RAP-x coatings by an optical microscope at 25 °C in air. The scale bar is 500 μ m. (d) PU-RAP-5.0 sample was healed at 25 °C for 24 h after cut into separated pieces and joined together.

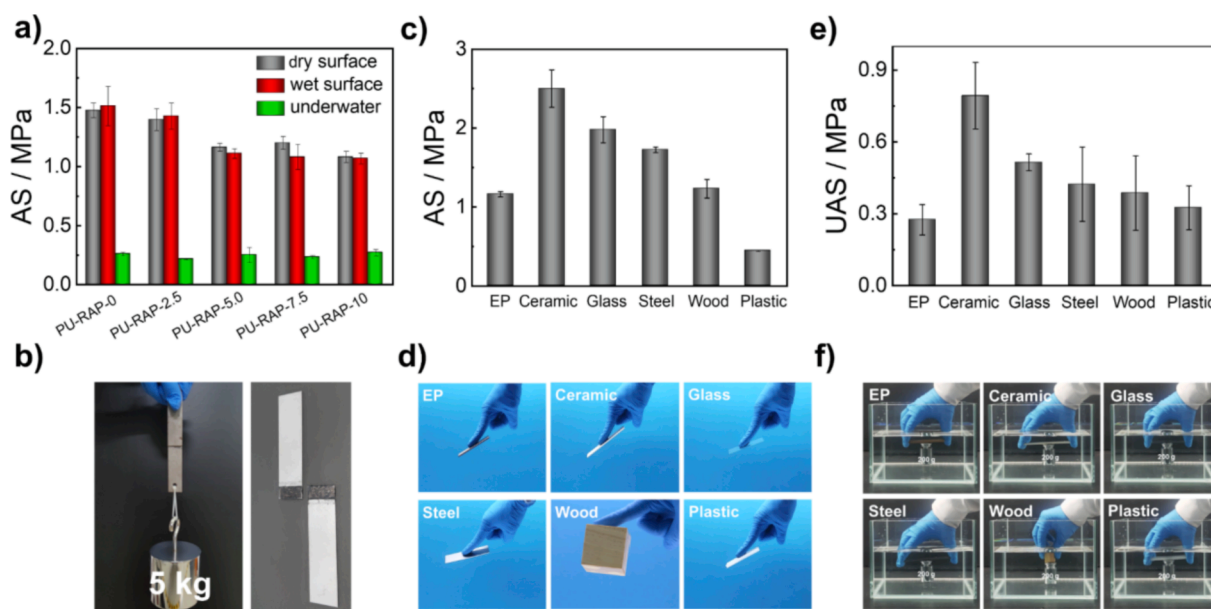


Fig. 3. (a) Adhesion strength (AS) of PU-RAP-x coatings under dry surface, wet surface, and underwater surface. (b) Steel lap joints adhered by PU-RAP-5.0 coating can support a weight of 5 kg and the cohesive failure of PU-RAP-5.0 coating on steel. (c) AS of PU-RAP-5.0 coating on various surfaces, such as epoxy panel (EP), ceramic, glass, steel, wood, and plastic. (d) Adhesion property of the PU-RAP-5.0 coating for attaching on various surfaces. (e) Underwater adhesion strength (UAS) of PU-RAP-5.0 coatings on various surfaces. (f) Underwater adhesion exhibitions of PU-RAP-5.0 coating on various surfaces to lift a steel weight of 200 g.

PTMG play critical roles in the adhesion strength [41]. Moreover, the dopamine and silane groups further improve the adhesion strength due to the formation of physical and chemical bonds in the interfaces even with a low content [42]. As shown in Fig. 3b, the steel lap joints adhered by the PU-RAP-5.0 coating can support a weight of 5 kg, indicating the strong AS of the coating. Furthermore, due to the formation of coordination bonds between Fe and catechol groups, the coating on steel has a black color (Fig. 3b) [43]. PU-RAP-x coatings also showed good adhesion properties to various surfaces. Fig. 3c and 3d show that the PU-RAP-5.0 coating has a high AS to the ceramic (~2.5 MPa), glass (~2.0 MPa), steel (~1.7 MPa), and wood (~1.2 MPa) in air. In addition to hydrogen bonds, the coating is probably covalently bonded to the

substrates through the reaction of pendant silane groups of PU with hydroxyl groups on ceramic or glass. For the wood, the high adhesion strength was due to the porous and rough surface of wood that allows a larger surface area to form hydrogen bonds [44]. Note that the coating has a low AS to plastic (~0.4 MPa). This is understandable because the plastic surface is nonpolar and too hydrophobic.

Although the bonding behaviors of many coatings reported before are excellent in dry conditions, the adhesion strength of the majority of these coatings is dramatically lower in wet conditions or invalid underwater [39,45]. We further evaluated the AS on wet surfaces and underwater (Fig. 3a). The AS of PU-RAP-x coatings on wet surface is similar to that of the dry surface. However, the underwater adhesion

strength (UAS) decreases to ~ 0.2 MPa on EP substrates. The decline of the UAS is understandable. On the one hand, the water on the interface forms a hydration layer, preventing molecular-level bridging between the coating and the substrate. Furthermore, water drops tend to get trapped at the interface, reducing the contact area macroscopically [27]. In addition, we examined the UAS of PU-RAP-5.0 coating on various surfaces. Fig. 3e shows that the measured UAS values of PU-RAP-5.0 on various surfaces are in the range of 0.2–0.8 MPa in water, which is better than the reported underwater adhesive materials [27]. To exhibit the underwater adhesion, the PU-RAP-5.0 sample and the different substrate were both immersed in water and then brought them together by gently pressing them against each other to ensure full contact (Fig. 3f). On different underwater substrates, PU-RAP-5.0 with a weight of 0.1 g can lift a weight of 200 g, indicating its excellent underwater adhesion. The high adhesion of PU-RAP-x coatings in variety of surfaces in both air and water is advantageous for biomedical and marine device application.

The anti-biofouling performance of the PU-RAP-x coatings was examined using proteins, bacteria, and diatoms as the typical fouling organisms. We firstly examined the protein resistance of PU-RAP-x coatings by using Quartz Crystal Microbalance with Dissipation (QCM-D) measurements. Fig. 4a shows the time dependences of frequency shift (Δf) and energy dissipation shift (ΔD) for the adsorption of fibrinogen on various PU-RAP-x surfaces, where fibrinogen is a large hydrophobic protein and widely used as a model [46]. For PU-RAP-0 surfaces, the remarkable decrease in Δf and increase in ΔD after the introduction of fibrinogen and rinsing indicate the strong protein adsorption on PU-RAP-0 surfaces. PU-RAP-2.5 also shows an obvious decrease in Δf and an increase in ΔD , indicating limited protein resistance. In contrast, only minor changes in Δf and ΔD are observed on PU-RAP-5.0, 7.5, and 10 surfaces after the introduction of fibrinogen solution, demonstrating exceptional protein resistance and indicating that protein resistance

improves with RAP content. It is well-known that PEG groups can form a hydration layer with their interfacial water molecule, which serves as a barrier to effectively resist protein adsorption [14,47,48]. As discussed above, the RAP contain PEG groups can migrate to the surfaces of coating, leading to protein resistance.

To evaluate the antibacterial properties of PU-RAP-x coatings, the antibacterial assays were conducted using two representative bacteria, *E. coli* and *S. aureus*, which are widely used in our daily environment. As shown in Fig. 4b and 4c, after 4-h exposure in bacterial suspensions, there are high-density of *E. coli* colonies on the control (~ 2000 cfu $\cdot$ cm $^{-2}$) and PU-RAP-0 surfaces (~ 1500 cfu $\cdot$ cm $^{-2}$). In contrast, only a small number of bacterial colonies could be observed on the PU-RAP-2.5 (~ 500 cfu $\cdot$ cm $^{-2}$), PU-RAP-5.0 (~ 75 cfu $\cdot$ cm $^{-2}$), PU-RAP-7.5 (~ 15 cfu $\cdot$ cm $^{-2}$), and PU-RAP-10 (~ 17 cfu $\cdot$ cm $^{-2}$) surfaces. These coatings significantly prevent the adhesion of bacterial and the antibacterial efficiency increases with the RAP contents. The antibacterial properties toward *S. aureus* are also confirmed by Fig. 4b and 4d. The antibacterial assay by fluorescent method and its relative bacterial adhesion (RBA) are shown in Fig. 4b and Fig. S9. A similar trend is obtained when compared to the colonial plate method. We also evaluated the antibacterial properties of the PU-RAP-x coatings (Fig. S10) using the marine bacterium *Pseudomonas* sp. nov 776 (*P. sp.*) which is derived from real marine environments and widely used as the model to evaluate the coating for marine antifouling [49]. The same phenomenon is observed for *P. sp.* compared to the *E. coli* and *S. aureus*. A large number of bacteria are adhered on the surfaces of the control glass slide as well as the PU-RAP-0 ($\sim 87\%$ bacterial adhesion). As the RAP content increases, the adhesion of bacterial decreases, or the fouling resistance increases.

Diatoms are another representative fouling organisms that settle easily on submerged surfaces [50]. As shown in Fig. 4e and Fig. S11, after 24 h settlement, a greater density of diatoms (460 cell $\cdot$ mm $^{-2}$) can be observed in the surfaces of PU-RAP-0 compared with the control (480

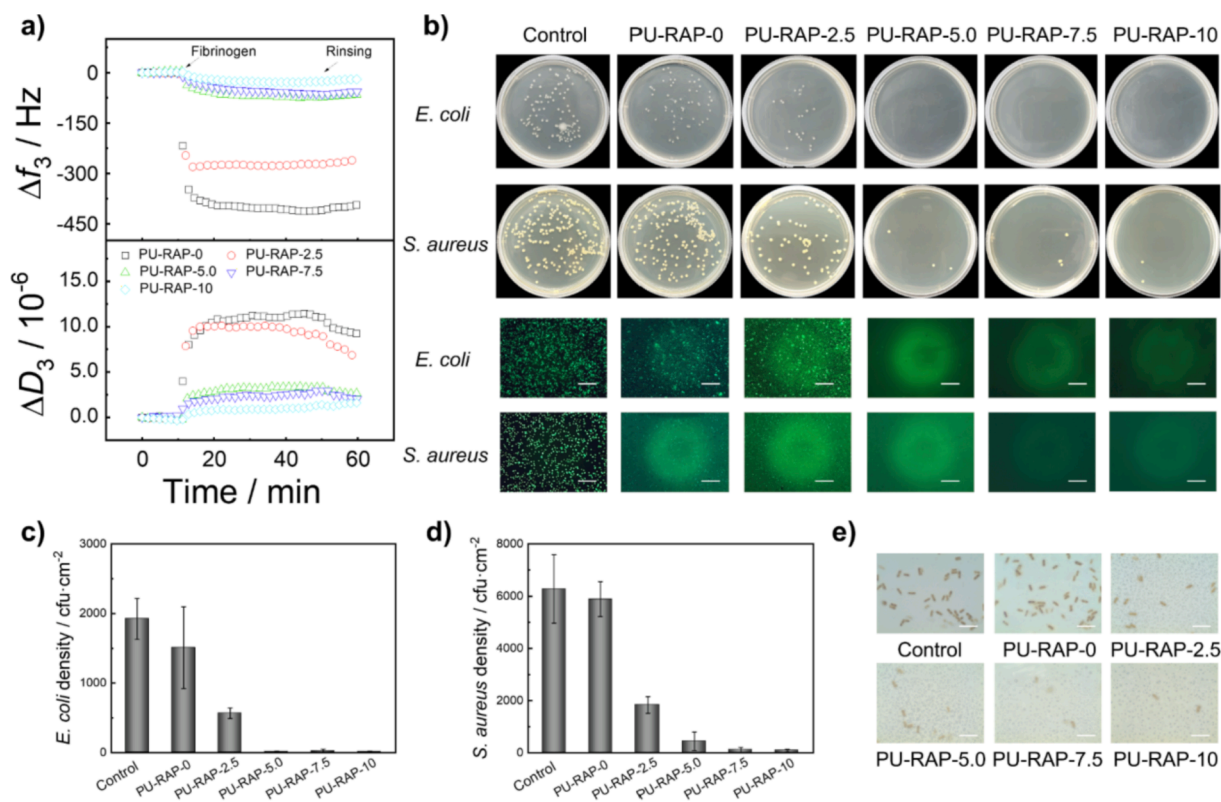


Fig. 4. (a) Change of frequency (Δf) and dissipation (ΔD) regarding adsorption of fibrinogen on PU-RAP-x coatings by QCM-D. (b) Colonial morphology images and fluorescence microscopy images of *E. coli* and *S. aureus* attached on PU-RAP-x coatings. The scale bar is 50 μ m. (c) and (d) Numbers of *E. coli* and *S. aureus* attached on PU-RAP-x coatings, respectively. (e) Microscope images of *Navicula incerta* diatom adhered on the PU-RAP-x coatings. The scale bar is 50 μ m.

cell-mm⁻²). After the introduction of the RAP, PU-RAP-x coatings show outstanding anti-diatom performance compared to the PU-RAP-0. Typically, the PU-RAP-2.5, 5.0, 7.5, and 10 coatings have nearly 140, 90, 45, and 30 cell-mm⁻² respectively, indicating that the adhesion diatoms decreased with an increase in RAP contents. The antifouling tests show that the coating has excellent antifouling performance even with a low content of amphiphilic polymers (~5 %), which is lower than that of the literatures reported [22,37].

4. Conclusion

In this study, we have developed a bioinspired self-stratifying strategy to simultaneously realize antifouling, self-healing, and strong substrate adhesion for a coating. The polyurethane with hydrogen bonds, pedant dopamine and silane groups in such coating systems imparts excellent adhesion strength to diverse surfaces, even in wet and underwater environments. Besides, the coating also shows good self-healing properties at room temperature because of the hydrogen bonds. Particularly, the amphiphilic segments can migrate to the surface of the coating during film formation due to its low surface energy, exhibiting outstanding antifouling abilities against proteins, bacteria, and diatoms even with a low content of reactive amphiphilic polymer. The coatings may find applications in marine antifouling, biomedical instruments, and other fields.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2022.137875>.

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